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A Crawler Climbing Robot Integrating Electroadhesion and Electrostatic Actuation

Regular Paper

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Abstract

Previous electroadhesive climbing robots generally employed typical electromagnetic motors, which spoiled some of the advantages of electroadhesion such as it being light, thin and flexible. To improve these, an integration of electrostatic actuation and adhesion was utilized in this work. By using FEM analyses, the present paper analysed the effect of design parameters on adhesive and driving forces respectively, and examined the possible interference between electrostatic actuation and adhesion inside the integration. Then this article discussed the driving force, payload capacity and torque balance of the robot with the integration. Based on these analyses, we designed and fabricated a lightweight (94 g) and low-height (15 mm) prototype with electrode films made by screen printing. Experiments on the prototype demonstrated that it can adhere to a vertical wall stably and move at a maximum speed of 35.3 mm/s.

Keywords Climbing robot, Electroadhesion, Electrostatic actuator

1. Introduction

A wall climbing robot is a promising technique for inspection and maintenance tasks, due to its unique capability for mobility. Various prototypes have been developed to act on ships [1], aircrafts [2], nuclear power plants [3], and high-rise buildings [4], areas which are dangerous or barely accessible for human beings [5]. Unlike ground-moving robots, climbing robots face various specific challenges. Two of these are adhering to various surfaces and materials, and being light in weight while maintaining a sufficient payload capacity so as to reduce power consumption and improve mobility [6].

These requirements, however, cannot be easily fulfilled with the traditional adhesion methods used for climbing robots, such as air suction, magnetic adhesion and gripping with claws. For example, a climbing robot with static air pressure can only adhere to smooth and non-porous surfaces [4, 7], and ones with dynamic air flow can be heavy and noisy due to bulky air pumps or fans [8]. The magnetic adhesive force is, although strong, only available on ferromagnetic materials [9]. Claws perform well on a tree or a brick wall, but will fail on a smooth surface such as a glass or painted wall [10, 11]. To implement a climbing robot that can adapt to a wider range of substrates while maintaining a light weight, several new adhesion methods have been proposed. One example is the gecko-inspired adhesive pad, which is composed of thousands of tiny fibres that can bear a heavy load with considerably small

size and low weight [12]. With the sticky force of soft elastomer, a palm-sized climbing robot in a crawler configuration can climb on different materials. [13]

This paper focuses on another promising adhesion method, which is electroadhesion. It works based on the attraction force between the opposite electric charges in an adhesive pad and a substrate [14-17]. The electroadhesive pad is thin, flexible and light-weight since it can be made of a thin plastic film with embedded electrodes. It can adhere not only to conductive substrates but also to insulating ones [14, 15]. In addition, robots with electroadhesion are expected, although not experimentally verified, to operate in extraordinary environments, such as vacuums, space and strong magnetic fields.

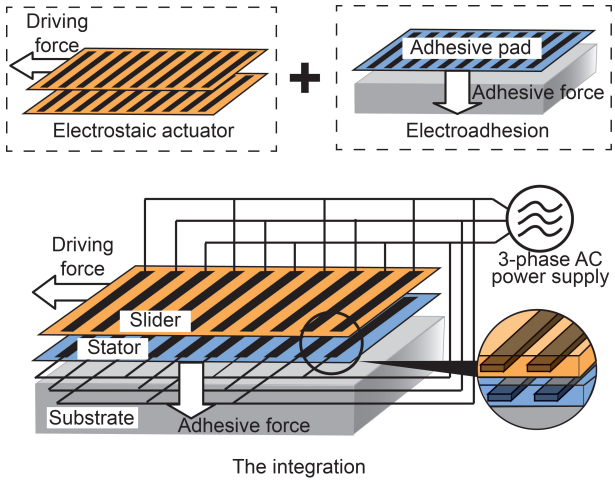


Figure 1. The integration of electrostatic actuation and adhesion

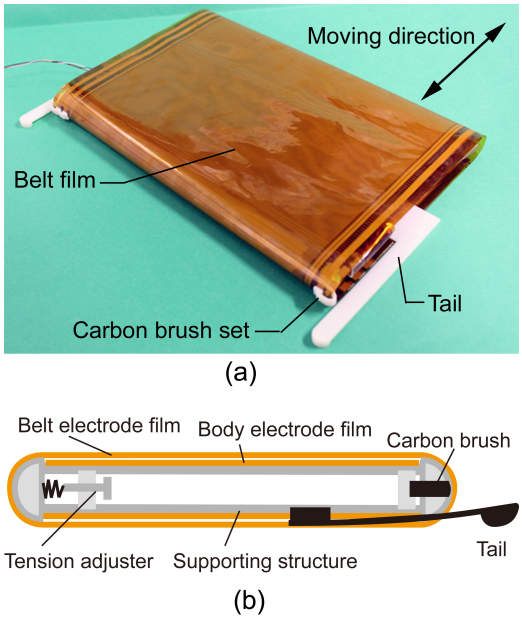


Figure 2. The appearance (a) and the structure (b) of the prototype of an electroadhesive crawler climbing robot driven by electrostatic actuators

The previously reported electroadhesive robots, however, cannot fully exploit these unique characteristics of electroadhesion, as they relied on electromagnetic motors for locomotion. For example, typical electromagnetic motors are heavy due to their ferromagnetic cores and metal wires, and their supplementary components for power transmission, such as dive belts and gearboxes, also take some weight. These components, including the electromagnetic motors, can occupy a noticeable percent of the robot weight, spoiling the light-weight features of electrostatic adhesive devices. In addition, the bulky electromagnetic motors prevent the robot from being compact, while the adhesive pad is only a thin film.

Therefore, it has been proposed in our previous work to replace the conventional electromagnetic motors with electrostatic actuators for electroadhesive climbing robots [16]. The electrostatic actuators employed in the proposal are generally made of thin electrode films [18-20], while electroadhesion usually relies on an electrode film of a similar structure. Therefore, the two force generation mechanisms can share one electrode film, as shown in Figure 1. The configuration depicted in this figure is referred to as "the integration" in the rest of this paper. Some of the advantages of the integration on a climbing robot are as follows:

- It simplifies the structure for the locomotion of the robot, because the electrostatic actuators can directly rotate the crawler belt film of the robot without any transmission mechanism, such as a gearbox and a drive belt.
- It allows the robot to be more compact due to the thin-thickness of the films used in both the electrostatic actuation and adhesion.
- The robot can be mechanically flexible, as both electrostatic technologies are based on thin and flexible plastic films.
- The robot can work in unique environments, such as vacuums and a strong magnetic field where magnetic motors are not qualified.

This previous work has demonstrated the primary prototypes of climbing robots that integrated electrostatic adhesion and actuation [16]. One of the prototypes was light in weight (49 g) and moved at a speed of 104 mm/s. The other prototype demonstrated structural flexibility that allows the robot body to comply with curved surfaces. However, these prototypes have the following problems.

- They could only move within a limited stroke, because the feeding wires, which are directly connected to the belt film, would be wound up as the robot moves.
- The size of the electrode films used in the robot was small due to the limitation of the manufacturing process and thus required connecting two films to form the crawler belt. This spoiled the locomotion ability as the connecting sections on the belt produced considerably large friction.

- The previous prototype was apt to be peeled away from a vertical substrate, especially when a disturbance happened.
- The previous report lacks theoretical analysis on the force generation mechanisms in the integration of actuation and adhesion, as well as on the force balance of the robot.

In this work, we theoretically analysed the climbing robot in terms of the integration of the electrostatic actuation and adhesion, and designed a new prototype (as shown in Figure 2) that overcomes the problems mentioned above. The structure of this paper is as follows. Section 2 briefly reviews the basic principles of the electrostatic actuation and adhesion, as well as their integration configuration in the robots. Section 3 analyses the effect of electrode designs on the performance of electroadhesion and electrostatic actuation, and examines the possible interference between the two force generation mechanisms. Section 4 analyses the force balances on the robot, including the driving force and the maximum payload of the robot, as well as the torque balance of the robot. Based on the analyses, a prototype was designed and fabricated, described in Section 5. Finally, experiments are conducted in Section 6 to verify the analyses, as well as to investigate the basic performance of the prototype climbing robot.

2. Basic principles of force generation mechanisms

2.1 Electrostatic actuation

This work employs the electrostatic actuation principle called *dual excitation multiphase electrostatic drive* [18]. Typically, the actuator includes two identical plastic films with embedded three-phase electrodes, as shown in Figure 1. A three-phase voltage supplier charges the electrodes in the two films with reversed connection orders. The resulting electrostatic interaction between electrodes actuates the slider in the tangential direction.

The synchronous speed of the slider, v, is calculated as

$$v = 6pf \tag{1}$$

where p is the pitch of electrodes, and f is the frequency of the applied three-phase voltage [21].

Its driving force, F_{dr}, depends on the slider position and the applied voltage. If we ignore force ripples, the force is expressed as [21]

$$F_{dr} = \frac{3\pi}{2p} C_a V^2 \sin\left(\theta_x - 2\varnothing + \pi/3\right) \tag{2}$$

where V is the zero-to-peak amplitude of the applied three-phase voltage, C_a is the variation of the capacitance (C_m) between one phase of the stator and one of the slider, θ_x is the slider position in electrical angle (in which three pitches of the slider displacement corresponds to 2π), and $\varnothing$ is the

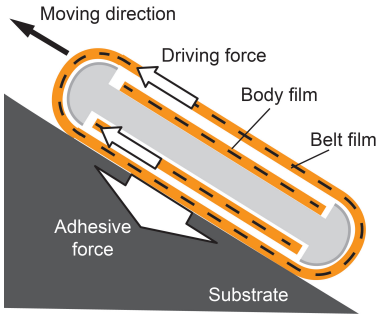


Figure 3. Basic mechanism of the electroadhesive crawler climbing robot driven by electrostatic actuators

voltage phase. The driving force is maximized when the argument of the sine function reaches $\pm\pi/2$. This maximum driving force will be mainly discussed in the simulation and experiments described later. Here we approximate that the capacitance changes between the stator and the slider as

$$C_m = C_0 + C_a \sin(\theta_x) \tag{3}$$

where C_0 is the average capacitance [21].

Since the electrostatic interaction also causes a vertical attraction force, we scattered small glass beads between the two films to reduce the friction force during operations. In some previous works on this actuator, an insulating fluid was injected into the gap between the films to increase the critical voltage of the discharge [16, 21], but in this work, the liquid was not utilized due to its difficulty with respect to maintenance, and hence, to avoid the corona discharge, the voltage was limited below 600 $V_{0\text{-}p}$.

2.2 Electroadhesion

Electrostatic adhesion generally employs a flexible adhesive pad with parallel electrodes inside. Previous studies possessed two-phase electrodes and activated them with a pair of DC voltages [14]. This research, however, proposed to utilize AC-excited three-phase electrodes. One of the advantages of three-phase electrodes is that it allows the adhesive pad to be used as an electrode film of the electrostatic actuators mentioned before. The other advantage is that three-phase electrodes can generate a constant adhesive force even if they are connected to three-phase AC voltages, while, in two-phase electrodes, the adhesive force oscillates when excited by AC voltages.

While calculating the normal adhesive force of electrostatic adhesion, we should consider the existence of an air gap between the contact surfaces, due to the intrinsic deformation of the electrode film and surface roughness. The adhesive force can be calculated by, [16]

$$F_{ad} = \frac{1}{8}\varepsilon_0 \varepsilon^2 S\left(\frac{V}{\varepsilon_0 t + \varepsilon t_0}\right)^2 \tag{4}$$

where ε and t are the permittivity and thickness of the dielectric layer respectively, ε_0 and t_0 are the permittivity and distance of the air gap, and S is the total area of the electrode film. Note that the electrode width is assumed to be half of the electrode pitch.

2.3 The integration of the electrostatic actuation and adhesion for a crawler climbing robot

The basic idea of the integration of electrostatic actuation and adhesion is to utilize the stator of an electrostatic actuator as an adhesion pad at the same time, as depicted in Figure 1. The integration can be employed in a crawler climbing robot with the configuration shown in Figure 3 [16].

The robot utilizes three electrode films: two short films attached to the top and bottom surfaces of the supporting structure, and one long film that works as a belt. Hereafter they are referred to as "body (electrode) films" and "belt (electrode) film" respectively. The belt film is driven by the body films to rotate while the bottom section of the belt film is adhering to the substrate.

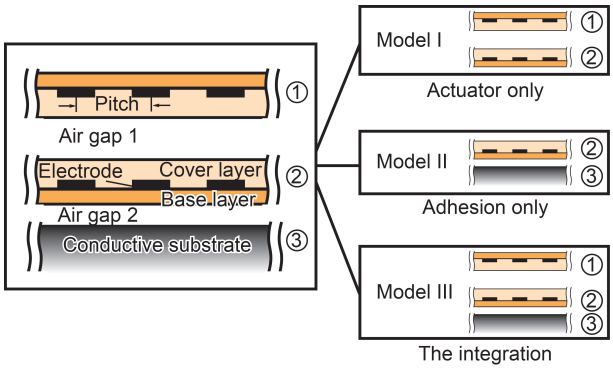


Figure 4. Sectional views of the simulation models for the analyses of electrostatic actuation and adhesion in their separated conditions (model I and II) and integration configuration (model III). The whole model has four sets of three phase electrodes for the upper electrode film and six for the lower one to reduce the fringing effect.

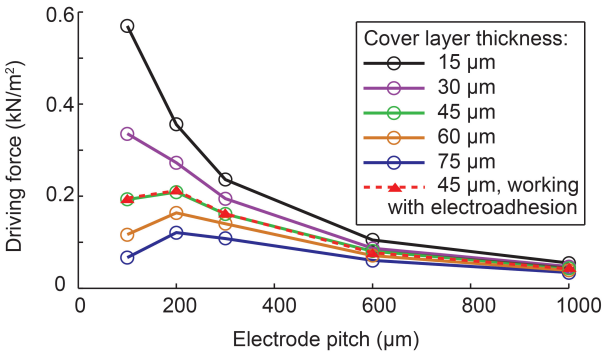


Figure 5. The simulation results of the driving force of an electrostatic actuator in the independent state, and in the integration (in which 45-μm cover layer is taken as an example, depicted by the red slashed line with triangle marks)

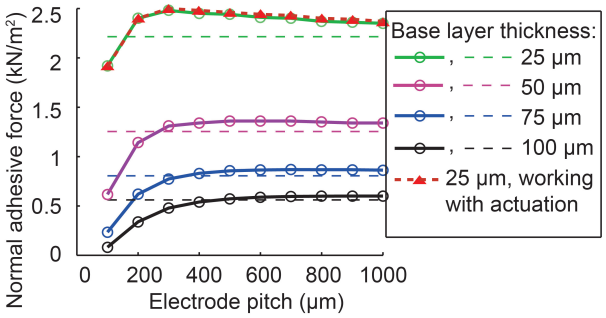


Figure 6. The adhesive force from the simulation model, compared to the results from equation (4) based on the parallel-plate capacitor model. When the adhesion works in the integration, accompanying with the electrostatic actuation, the adhesive force of the electrode film owning the 25-μm base layer is also shown for comparison (the red slashed line with triangle marks).

3. Force generation mechanisms in the integration

Using three independent models, this section discusses the driving and adhesive forces, and the possible interference between actuation and adhesion in the integration presented in Figure 1.

3.1 The simulation models

This work built three FEM models as shown in Figure 4. The first model (model I) describes an actuator alone and contains two overlapping films. The second model (model II) describes electroadhesion alone that includes only one film on a substrate. The third model (model III) describes the integration model that combines the two films and the substrate together. Assuming the films in the models are made by screen printing, each film has a base layer, a cover layer, and several coplanar electrodes sandwiched inside.

To facilitate the design of the electrode films, we examined the effects of three main parameters — the pitch, and the thicknesses of the base and cover layers, which were selected with the limitation of screen printing and for the simplicity of analysis. Note that the thickness of the cover layer refers to the distance from the electrodes to the surface of the cover layer. The electrode width is fixed to a half of the electrode pitch. The relative permittivity of the cover layer and base layer are assumed to be 2.24 and 3.4, respectively. Two air gaps are assumed: one is between the two films and is 50-μm thick (including the diameter of glass beads, 20 μm), while the other is between the film and substrate and is 5 μm. The amplitude of the three-phase voltage is 1 $kV_{0\text{-}p}$.

3.2 Electrostatic driving force

Using model I shown in Figure 4, we calculated the driving force for different thicknesses of the cover layer while the base layer thickness remains constant at 25 μm. The force calculation was carried out indirectly to fit into the approximations given in equations (2) and (3). First, we obtained the capacitance C_m (θ_x) by shifting the slider over three pitches with several steps. The capacitance obtained at this

step has some distortion which corresponds to the force ripple. As we neglect the force ripple for easier comprehension of the performance, we removed the distortion by transforming C_m into the frequency domain using the Fast Fourier Transformation, and extracted the amplitude of the first component of C_m as the approximate value of the characteristic capacitance C_a. Finally, the driving force was calculated using the obtained C_a and equation (2).

Figure 5 plots the driving force with respect to pitches at different thicknesses of the cover layer. Generally, the driving force increases as the thickness of the cover layer or the pitch decreases. However, the driving force descends for thick insulation (for example, 75 μm) and small pitches (smaller than 200 μm). This would result from the small pitch/gap ratio which reduces the capacitance variation with regard to the position change.

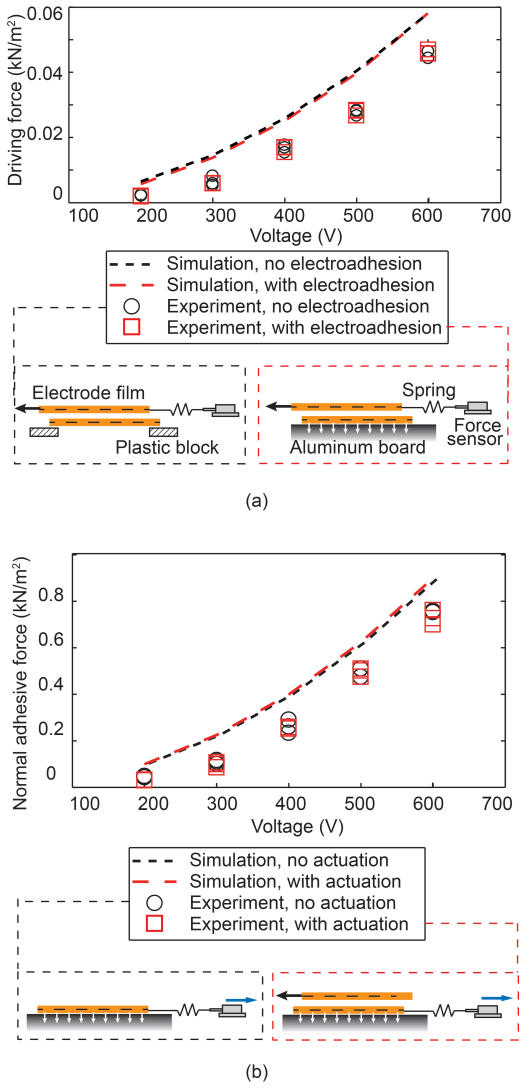


Figure 7. The driving force (a) and the adhesive force (b) in the integration, compared with their counterparts in the independent configuration of actuation or adhesion

3.3 Electrostatic adhesive force

In the simulation of electroadhesion, a three-phase voltage ($\sin(\phi)$,$\sin(\phi+2\pi/3)$,$\sin(\phi-2\pi/3)$) was supplied to the three-phase electrodes in model II, and a normal electrostatic force that acts as an adhesive force was calculated. As we did not observe a noticeable dependence of the normal adhesive force on the phase ϕ, the phase was kept at $\pi/2$ in the following simulations. We changed the electrode pitch and base layer thickness but kept the thickness of the insulation layer to 45 μm. The simulated results for the different parameters are shown in Figure 6. This figure compares the simulated results with the calculated results of equation (4) which is based on the parallel-plate capacitor model. Similar to the calculated results, the simulated adhesive force stays almost constant at large electrode pitches. At small pitches, however, the simulation results degrade probably resulting from the electric field distortion caused by the closer distance to nearby electrodes.

3.4 The possible interference between electrostatic actuation and adhesion in the integration

To investigate the interference between electrostatic actuation and adhesion in the integration, we analysed both the adhesive and driving forces in the integration model (model III shown in Figure 4), which was compared with the results obtained in the separated models (model I and II).

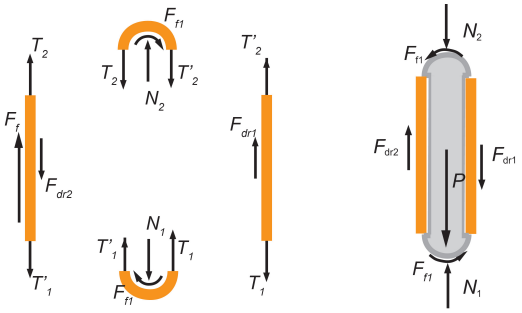


Figure 8. The forces on the belt film and the supporting structure

As presented in Figure 5 and Figure 6, the driving and adhesive forces in the integration almost perfectly match the results obtained from the separated conditions. This suggests that the interference between electrostatic actuation and adhesion is negligible. Accordingly, we can directly employ the data of the separated models into the integration for the design of robots, which is beneficial to simplifying the design process.

3.5 Experimental verification

The analyses on the driving and adhesive forces were also verified by experiments on the driving and adhesive forces respectively. Figure 7(a) compares the tested driving forces in two cases. In one case, the actuator worked alone without any adjacent metallic object; in the other case, an alumini-

um board was located under the stator of the actuator, which means electroadhesion was also involved in this case. The electrode films used in the experiments had the same structure, as will be depicted in detail in Section 5. The forces measured for the two situations match almost perfectly. This verifies that the electroadhesion does not affect the actuation, as analysed above. Although some discrepancy can be found between the simulation results and the experimental data, the discrepancy falls within an acceptable range.

As for the adhesive force, its experimental results and the setups are all shown in Figure 7(b). In the experiments, the friction force caused by the adhesive force was measured, which was then converted to normal adhesive force using the pre-identified friction coefficient. The adhesive force was measured in two different cases. In one case a single film containing three-phase electrodes was placed on an aluminium board; in the other case, two films that comprise a driving actuator was placed on the board. The forces were found almost identical for the two cases and fairly match the simulated results. Again, the results confirmed that electroadhesion and electrostatic actuation work independently in the integration.

4. Force analysis on the robot

4.1 Driving force and payload capacity

The forces on the robot adhering to a vertical surface can be analysed as shown in Figure 8. In the static/quasi-static state, the forces on the straight parts of the belt balance as,

$$T_1' + F_{dr2} = T_2 + F_f \tag{5}$$

$$T_1 = F_{dr1} + T_2' \tag{6}$$

in which T_1, T_2, T_1' and T_2' are the forces on the belt film, while F_{dr1} and F_{dr2} are the driving forces, as shown in Figure 8.

The force balance on the curved parts of the belt is

$$k = \frac{T_1}{T_1'} = \frac{T_2}{T_2'} \tag{7}$$

$$T_1 + T_1' = N_1 \tag{8}$$

$$T_2 + T_2' = N_2 \tag{9}$$

where k is a constant relative to the belt friction, and P is the payload on the robot (including the dead weight of the robot), while N_1 and N_2 are the normal supporting forces generated from the supporting structure.

We also have the following relationship,

$$N_1 = 2F_{pre} \tag{10}$$

$$N_2 = 2F_{pre} + N_B \tag{11}$$

where F_{pre} is the preloaded tension force given from the supporting structure to the belt film, and N_B is the extra force due to the payload.

The maximum payload that the robot can carry is determined either by the actuators' driving force or by the friction force on the substrate created by the electroadhesion. Ideally, if the two electrostatic actuators generate the same forces, (F_{dr1}=F_{dr2}=$F_{dr\text{-}max}$), the maximum payload P_{max} is calculated as:

$$P_{max} = \begin{cases} P_{dr\text{-}max} = \frac{k+1}{k} F_{dr\text{-}max} - \frac{2(k-1)}{k} F_{pre} \\ \left(P_{dr\text{-}max} < F_{f\text{-}max}\right) \\ F_{f\text{-}max} = \mu F_{ad} \\ \left(P_{dr\text{-}max} \geq F_{f\text{-}max}\right) \end{cases} \tag{12}$$

where $F_{dr\text{-}max}$ is the maximum output force which can be generated from one actuator. In the equation above, $P_{dr\text{-}max}$ expresses the maximum payload defined only by the actuators' driving force, and $F_{f\text{-}max}$ expresses the maximum friction force.

4.2 The effect of electrode misalignment on the payload

In equation (12), we have assumed that the two actuators generate the same thrust force. Since the two actuators are synchronous actuators, they must share the same slider position (relative to the corresponding stator) to generate the same thrust force under the same voltage. In practice, however, an electrode misalignment between the two actuators can happen as shown in Figure 9. Figure 9 illustrates a misaligned situation, where the lower actuator has a slider position of θ_x (in the electrical angle), whereas the upper actuator has θ_x+$\Delta\theta_x$; there is an electrode misalignment of $\Delta\theta_x$. In this situation, according to equation (2) and (5)-(10), the maximum payload defined only by the actuators' driving force can be calculated as

$$P_{dr\text{-}max} = \sqrt{1 + \frac{2}{k}\cos\Delta\theta_x + \frac{1}{k^2}} F_{dr\text{-}max} - 2\frac{k-1}{k} F_{pre} \tag{13}$$

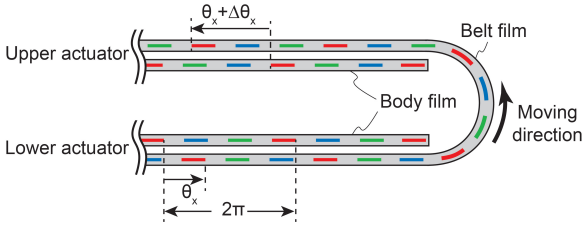


Figure 9. The misalignment in the two actuators

This equation reveals that the payload is significantly affected by the electrode misalignment between the two actuators. In Section 6, the payload of the prototype robot will be shown as an example for this situation.

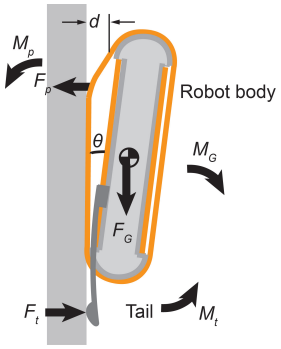


Figure 10. The balance of the crawler climbing robot on a vertical substrate, relying on the tails

4.3 Rotational torque balance on a vertical surface

A peel adhesive force refers to the bond strength that works between a film and a substrate, when the film is being pulled apart from the substrate. When the robot is peeling off from a vertical wall by a disturbance or by its own weight, the peel adhesive force underneath the head of the robot is expected to pull the robot back to the original posture, as shown in Figure 10. However, the electrostatic peel adhesive force of the robot appears not to be strong enough to stabilize the robot, as verified later in Section 5. Therefore, other structures that can exert a balance force should be introduced to prevent peeling. Tails are a good choice as studied in [13]. In this work, we also attached tails to the robot such that the total recovering torque generated by the peel adhesive force and the tails always surpass the gravitational torque.

5. Design and fabrication of the robot

The new prototype of the crawler climbing robots is shown in Figure 2. Its size and weight are shown in Table 1. The components acting for the actuation and adhesion in the climbing robot weigh only 21.4 g, because they are all made of thin plastic films. An advantage of this prototype over previously reported electroadhesive robots is found in the aspect of the ratio of weight per adhesive area. It weighs only 3.1 kg/m^2, much lighter than the other electroadhesive climbing robots which weigh at least 7.5 kg/m^2 as shown in Table 1. As the electroadhesive force depends on the area, the small weight/area ratio of the present prototype allows its exciting voltage to be as low as 0.6 $kV_{0\text{-}p}$. At this voltage, the present prototype can carry the extra payload of about 0.4 N. As a comparison, if the voltage is also 0.6 $kV_{0\text{-}p}$, the robot described in [14] will fall down, because its friction force will drop to only 1.6 N (considering the adhesion force is proportional to the square of the voltage), which cannot even support the weight of the robot itself.

The prototypes	[14] (Yamamoto A)	[15] (Prahlad H)	[17] (Liu R)	The present prototype
Velocity (mm/s)*	6.6	150	0.2	35.3
Working voltage	1.5 $kV_{0\text{-}p}$ (two-phase AC)	4 kV (DC)	3 kV (DC)	0.6 $kV_{0\text{-}p}$ (three-phase AC)
Friction force from electroadhesion (N)	10 N (on a surface-insulated conductive wall)	--	20 N (on a drywall)	6.68 N (on an aluminium board)
Dimension (length×width)	300 mm×130 mm	--	360 mm×360 mm	173 mm×274 mm (height: 15 mm)
Effective adhesive area (m^2)	0.04	0.024**	0.086**	0.03
Weight (g)	327	180	700 (with on-board power)	94
Weight per adhesive area (kg/ m^2)	8.2	7.5**	8.1**	3.1

*The prototype in [14] is inchworm type, while the others are all crawler type

**Estimated value

Table 1. The parameters of the prototype of electrostatic adhesive climbing robot driven by electrostatic actuators, compared with previously developed electroadhesive climbing robots driven by electromagnetic motors

The structure of this prototype is basically the same as the previously reported proof-of-concept prototype in [16] except for some improvements as follows. The electrode films for the new robot were fabricated using screen printing since it can implement a larger film with a higher structural flexibility, whereas the films in [16] were fabricated by the flexible printed circuit (FPC) technology that has a size limitation. Bare bus lines are added to the

new belt film to allow a voltage application through carbon brushes, which can avoid the twisting of the feeding wires. Tails were also newly installed on the climbing robot for the rotational stability. The details are explained as follows.

5.1 The electrode films

Based on the analyses above, we chose the parameters of electrode films such as the pitch and the thicknesses of the base layer and the cover layer. The pitch was selected at 300 μm. At this pitch, the driving force has a reasonable value for the robot. If the electrode pitch further reduces, the adhesive force will be weakened, and the manufacturing difficulty will be increased. The base layer thickness was selected to be 25 μm such that the film has a strong adhesive force but without any difficulty in handling. Finally, we chose the thickness of the cover layer considering two aspects: if the cover layer becomes thinner, a risk of electric discharge will arise; on the other hand, if the cover layer is thicker, the driving force will be weakened. Therefore, considering the trade-off between the two aspects above, the thickness of the cover layer was compromised to be about 45 μm.

All the electrode films used in this paper were made in the same structure as shown in Figure 11 by screen printing. We printed linear electrodes with a 300 μm pitch using a silver paste, on a base film of polyimide, which is covered by an insulation resin that functions as the cover layer. Finally, the electrodes were connected by three bus lines through holes in the cover layer. These bus lines were not covered by the cover layer to facilitate the voltage feeding by carbon brushes.

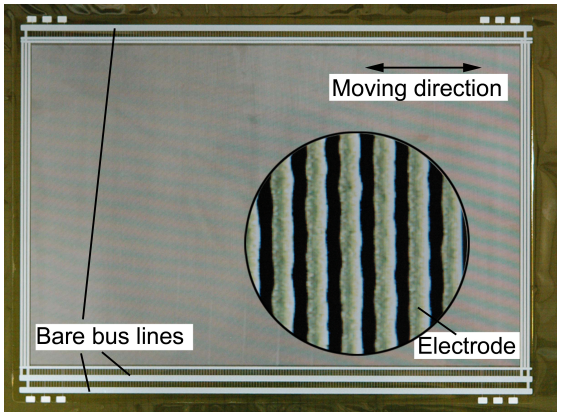


Figure 11. The electrode printed film manufactured by screen printing and used for the prototype (the electrode pitch is 300 μm, and the thicknesses of the base layer and cover layer are 25 μm and 45 μm, respectively)

The belt film was made by connecting both ends of the film mentioned above by tape. The body film attached on the upper and lower surfaces of the supporting body were made by cutting sections of 135-mm width from the film.

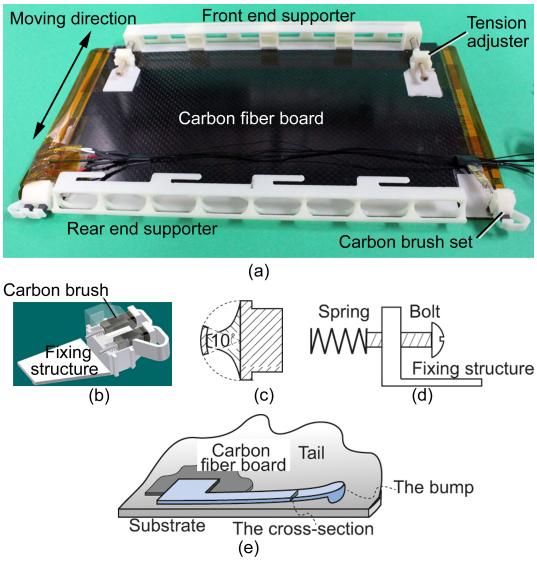


Figure 12. The components of the prototype, including (a) the carbon fiber board, (b) carbon brush set, (c) end supporter (cross-section), (d) tension adjuster and (e) tail, (the upper supporting plate (PET film) is not shown)

The bared bus lines on the body films were covered by tape in case of short circuit.

5.2 The mechanical structure of the robot

In conventional climbing robots, rollers are used to transmit the driving force generated from electromagnetic motors to the crawler belts. However, in the climbing robot of this paper, the driving force of electrostatic film actuators can drive the belt directly, so moving parts such as rollers are unnecessary. Therefore, it brings more freedom of design. In this case, we designed two fixed end supporters on the robot instead of rollers for weight reduction and simplicity.

As shown in Figure 12(a), one of the two end supporters is attached to an end of a rigid carbon fibre board (0.5 mm in thickness), and the other one is supported by tension adjusters on the other end of the same board. The tension adjusters as shown in Figure 12(d) can preload tension force to the belt film by pressing the front end supporter.

To feed three-phase voltage, the robot uses two three carbon brushes. The carbon brushes are contained in the fixing structures as shown in Figure 12(b) to ensure the close contact between the brushes and the bus lines on the belt film.

Two tails are symmetrically installed on the left and right sides of the carbon fibre board as they stick out from the rear end of the robot body, see Figure 2. When the robot is adhering to a substrate, the tails will be bent and generate a balance torque to the robot, as shown in Figure 12(e).

6. Experiments on the prototype

The prototype successfully climbed on the aluminium board as shown in Figure 13. The voltage required for the

successful operation was found in a range from 500 $V_{0\text{-}p}$ to 600 $V_{0\text{-}p}$ on a horizontal surface, while it was 550 $V_{0\text{-}p}$ to 600 $V_{0\text{-}p}$ on a vertical surface. It was confirmed that the robot can move continuously without stroke limitation owing to the power feeding by carbon brushes. In the experiments, no electric spark was observed between the carbon brushes and the bus lines on the film.

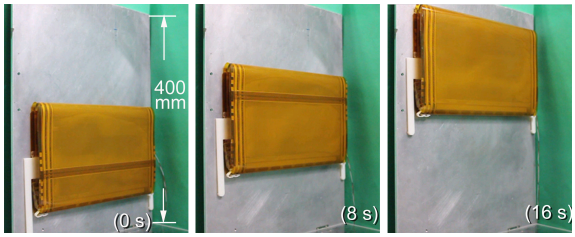


Figure 13. The snapshot of the prototype while it was climbing up a vertical aluminum board (5 Hz, 600 $V_{0\text{-}p}$)

6.1 Peel adhesive force of the robot

To clarify the magnitude of peel adhesive force under the head of the climbing robot, we conducted an experiment using the setup shown in Figure 14. The prototype robot (without the tails) was placed on a horizontal aluminium board. Its head was connected to a force sensor by a string. A pulley was used to keep the direction of the pulling force almost vertical during the peeling process. In the measurement, the force sensor was pulled back slowly to peel the robot off the aluminium board. Three-phase voltage of 600 $V_{0\text{-}p}$ was applied to the electrodes.

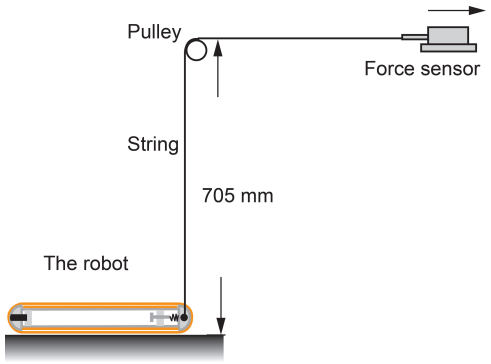


Figure 14. The setup for testing the peel adhesive force under the head of the robot

By excluding the effect of the robot's weight from the tested results, the peel adhesive force of the robot was estimated. As shown in Figure 15, it reached the maximum adhesive force of 94 mN, and then dropped to around 34 mN. The maximum peel adhesive force underneath the robot head can create balance torque of 12.7 Nmm, which seems enough to balance against the peeling off torque of gravity (6.4 Nmm). However, rotational stability provided by the peel adhesion is quite marginal, and once peeling process passes over the maximum adhesive force point due to a disturbance, the peel adhesive force will drop to one third of the maximum value. As the corresponding balance torque is only 4.6 Nmm. Therefore the robot on a vertical surface is not stable enough, if there is no tail.

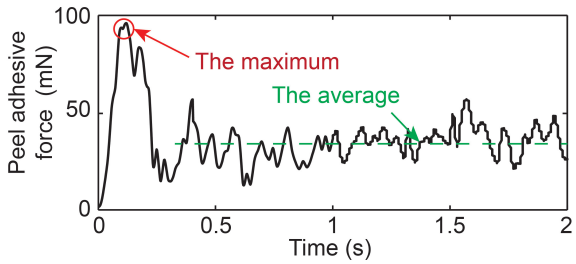


Figure 15. The peel adhesive forces underneath the head of the climbing robot

6.2 Friction force caused by the electroadhesion under the robot

To test the performance of the robot, the prototype was activated to adhere on a horizontal aluminium board. Its belt film was supplied with the three-phase voltage of 5 Hz. A force sensor horizontally pulled the robot through a spring to measure the friction force originated from electroadhesion. The results are shown in Figure 16.

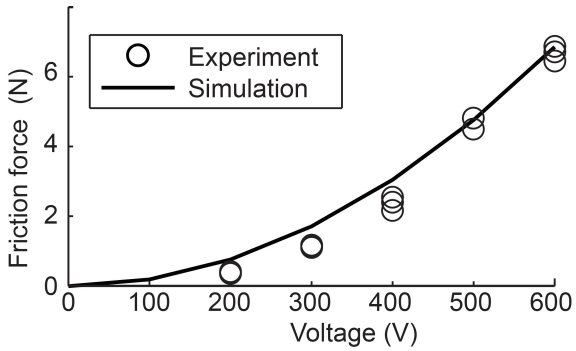


Figure 16. Maximum fiction force originated from electroadhesion underneath the climbing robot

As expected, the results of the friction force based on adhesive force are proportional to the square of the voltage. The maximum friction force was 6.68 N at the applied voltage of 600 $V_{0\text{-}p}$, which is enough to support the robot's weight of 94 g on a vertical wall.

6.3 The payload capacity of the prototype robot

As mentioned above, the friction force originated from the electroadhesion is strong enough to hold the robot on a vertical substrate, but it was observed that the robot could not carry a payload as great as the maximum friction force. When the payload of the prototype on a vertical wall increased, the robot actuation failed due to the step-out of the actuators before we observed the slip of the prototype on the substrate. This means that the robot's maximum payload was limited by its net driving force (in other words,

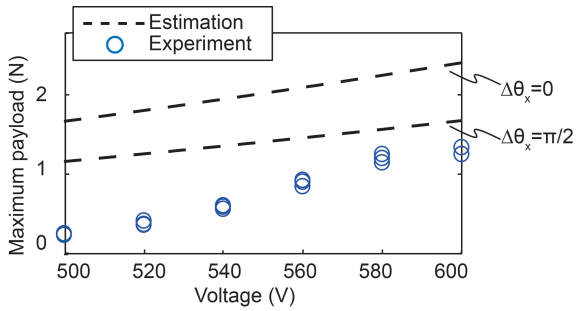


Figure 17. Payload capacity of the climbing robot

the first condition in equation (12) held in this particular situation). To test the payload capacity, we connected the robot to a force sensor through a spring, and drove it on a horizontal aluminium board to pull the sensor. The driving voltage was 5 Hz, which corresponds to the synchronous speed of 9 mm/s.

As shown in Figure 17, the maximum payload increased as the voltage increased. The measured payload reached 1.3 N at the voltage of 600 $V_{0\text{-}p}$, which was strong enough to drive the own weight of the prototype, 94 g, on a vertical surface. The payload is, however, only a half of the theoretical value calculated from equation (12), which is shown using a dashed line in the plot with a legend of "$\Delta\theta_x$=0". (Note that, for the calculation of the theoretical value, the driving force measured in Section 3 was used, F_{pre} was approximated to 0.5 N, and k was set to 1.06 based on a preliminary test.) This suggests that the prototype has an electrode misalignment between the two actuators.

In the prototype, the electrode pitch was merely 300 μm. Considering that the belt film is flexible, the misalignment was apt to happen, even though we had carefully installed the actuators to reduce it. The measured payload is rather close to the estimated value for the misalignment of π/2 from equation (13). This can be explained as follows. If there is play in the belt, the slider in the upper actuator can freely move within the play. In such a case, the slider tends to move to the equilibrium position, where the driving force of the upper actuator becomes zero, which results in the electrode misalignment of π/2. In this case, the upper actuator does not contribute to the payload. It is, however, still useful, because having actuators on both sides allows the robot to operate, no matter which side of the robot contacts the ground. This is beneficial for a recovery from a drop from a wall.

6.4 Velocity

We drove the robot on a vertical aluminium board and on a horizontal one respectively to measure its velocity. The amplitude of the three-phase voltage was kept at 600 $V_{0\text{-}p}$. The velocity was measured by a laser displacement sensor. Figure 18 shows that the recorded velocity almost matches the synchronous speed calculated by equation (1), which means no slip occurred underneath the robot. The robot could run as fast as 35.3 mm/s on a vertical surface in the test.

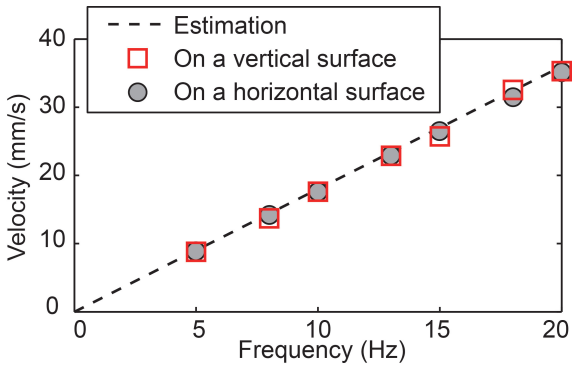


Figure 18. Velocity of the prototype of crawler climbing robot

7. Conclusion and future work

This paper studies a climbing robot with the integration of the two force generation mechanisms — electrostatic actuation and adhesion. To facilitate the electrode design, the paper analysed the effects of the electrode design on the electrostatic actuation and adhesion. The analysis revealed that the actuation and adhesion do not interfere in the integration, which was also confirmed by experiments.

The analysis of the mechanical force balance of the robot provided an equation to calculate the payload capacity of the robot. It was also revealed that an electrode misalignment between the two actuators, which are installed on both surfaces of the robot, can degrade the payload performance.

The experiments verified that the prototype can pull a payload of 1.3 N horizontally and move on a vertical wall with a maximum speed of 35.3 mm/s. The experimental results showed that there was an electrode misalignment and it was discussed that the misalignment was an inevitable problem for this type of robot. This problem can be solved by utilizing an asynchronous actuation method, since its driving force does not depend on the slider position. The asynchronous actuation proposed in [22] utilizes the same electrode structure as the synchronous actuator in this work, and therefore will be a good alternative actuator in future works. Another challenge that should be addressed in future works is to develop an on-board power supply so that the robot can operate without any power feeding cables.

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